

Lecture 2: Log Linearization

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Business cycle is fluctuation / deviation around trend. Log linearization: take logs to linearize the system and then take deviation from the trend / steady state.

* Can not work with discontinuous functions, eg: zero lower bound, default, work choice, etc.

1. Taylor expansions

Taylor expansion of order k of function f around a is to derive the best approximation of f around a by a polynomial of order k .

Suppose f is $n + 1$ times differentiable in an interval around a

$$x \in (a - \varepsilon, a + \varepsilon),$$

then

$$f(x) = f(a) + \sum_{i=1}^n \frac{f^{(i)}(a)}{i!} (x-a)^i + \underbrace{o((x-a)^n)}_{\text{Peano remainder terms}}.$$

eg:

$$\text{1st order: } f(x) \approx f(a) + \frac{f'(a)}{1!} (x-a)^1 = f(a) + f'(a)(x-a)$$

$$\text{2nd order: } f(x) \approx f(a) + \frac{f'(a)}{1!} (x-a)^1 + \frac{f''(a)}{2!} (x-a)^2 = f(a) + f'(a)(x-a) + \frac{f''(a)}{2} (x-a)^2$$

2. Taylor expansion on 2 dimensions

Suppose f is second-order continuous (C^2) in a circle around (x_0, y_0) that contains $(x_0 + h, y_0 + k)$.

For a small move (h, k) around (x_0, y_0) , how to approximate changes in f .

$$f(x_0+h, y_0+k) = f(x_0, y_0) + \left[f_1(x_0, y_0)h + f_2(x_0, y_0)k \right] + \frac{1}{2} \left[f_{11}(x, y)h^2 + 2f_{12}(x, y)hk + f_{22}(x, y)k^2 \right]$$

where

$$(x, y) = (x_0 + ch, y_0 + ck) \quad \text{for some } c \in (0, 1).$$

A vector format:

$$f(z_0 + \Delta) = f(z_0) + \nabla f(z_0)^T \Delta + \frac{1}{2} \Delta^T H_f(z_0 + c\Delta) \Delta$$

$$\text{where } z_0 = (x_0, y_0), \quad \Delta = (h, k)$$

3. Log linearization

percentage deviation: $\hat{x}_t = \log X_t - \underbrace{\log \bar{X}}_{\text{Steadystate}}$

Since $Y_t \rightarrow 0$ when Y_t is sufficiently small, we have

$$e^{Y_t} = e^0 + e^0(Y_t - 0) = 1 + Y_t \quad \Rightarrow \quad Y_t \approx \log(1 + Y_t).$$

$$\hat{x}_t = \log \frac{X_t}{\bar{X}} = \log \left(1 + \frac{X_t - \bar{X}}{\bar{X}} \right) \approx \frac{X_t - \bar{X}}{\bar{X}}.$$

$\hat{x}_t \times 100\%$ is approximately the percentage deviation w.r.t. S.S.

$$X_t = \bar{X} \left(\frac{X_t}{\bar{X}} \right) = \bar{X} e^{\log(X_t/\bar{X})} = \bar{X} e^{\hat{x}_t}$$

Taking first-order Taylor approximation:

$$\bar{X} e^{\hat{x}_t} \approx \bar{X} e^0 + \bar{X} e^0 (\hat{x}_t - 0) = \bar{X} + \bar{X} \hat{x}_t = (1 + \hat{x}_t) \bar{X}$$

$$\Rightarrow \quad X_t = \overbrace{\bar{X}}^{\text{Steady state}} + \overbrace{\bar{X} \hat{x}_t}^{\text{deviation from steady state}}$$

4. Examples

(1)

$$Y_t = a X_t$$

$$X_t \approx \bar{X}(1 + \hat{x}_t) \quad \Rightarrow \quad Y_t \approx a \bar{X}(1 + \hat{x}_t) = a \bar{X} + a \bar{X} \hat{x}_t$$

Since

$$Y_t = a X_t, \quad \forall t \quad \Rightarrow \quad \bar{Y} = a \bar{X},$$

we get

$$Y_t \approx \bar{Y} + \bar{Y} \hat{x}_t \quad \Rightarrow \quad Y_t \approx \bar{Y} + \bar{Y} \hat{y}_t$$

hence

$$\hat{x}_t = \hat{y}_t.$$

(2)

$$Z_t = X_t Y_t$$

$$X_t \approx \bar{X}(1 + \hat{x}_t), \quad Y_t \approx \bar{Y}(1 + \hat{y}_t)$$

$$\Rightarrow \quad Z_t \approx \bar{X} \bar{Y} (1 + \hat{x}_t)(1 + \hat{y}_t) = \bar{X} \bar{Y} \left(1 + \hat{x}_t + \hat{y}_t + \underbrace{\hat{x}_t \hat{y}_t}_{\approx 0} \right)$$

So

$$Z_t \approx \bar{X} \bar{Y} (1 + \hat{x}_t + \hat{y}_t).$$

Since

$$Z_t = X_t Y_t, \quad \forall t \quad \Rightarrow \quad \bar{Z} = \bar{X} \bar{Y},$$

we get

$$Z_t \approx \bar{Z} + \bar{Z}(\hat{x}_t + \hat{y}_t) \quad \Rightarrow \quad Z_t \approx \bar{Z} + \bar{Z} \hat{z}_t$$

hence

$$\hat{z}_t = \hat{x}_t + \hat{y}_t.$$

(3) Log linearize equation

$$X_t Y_t = C \quad \Rightarrow \quad \bar{X} \bar{Y} = C.$$

By (2),

$$X_t Y_t \approx \bar{X} \bar{Y} (1 + \hat{x}_t + \hat{y}_t) \quad \Rightarrow \quad C \approx C (1 + \hat{x}_t + \hat{y}_t)$$

so

$$\hat{x}_t + \hat{y}_t = 0.$$

(4)

$$\frac{X_t}{Y_t} = C \quad \Rightarrow \quad \frac{\bar{X}}{\bar{Y}} = C.$$

$$\frac{X_t}{Y_t} \approx \frac{\bar{X}(1 + \hat{x}_t)}{\bar{Y}(1 + \hat{y}_t)}$$

Use

$$(1 + \hat{y}_t)(1 - \hat{y}_t) = 1 + \hat{y}_t - \hat{y}_t - \hat{y}_t \hat{y}_t \quad \Rightarrow \quad \frac{1}{1 + \hat{y}_t} \approx (1 - \hat{y}_t).$$

Therefore

$$\frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}} (1 + \hat{x}_t)(1 - \hat{y}_t) \approx \frac{\bar{X}}{\bar{Y}} (1 + \hat{x}_t - \hat{y}_t - \hat{x}_t \hat{y}_t)$$

So

$$\frac{X_t}{Y_t} \approx \frac{\bar{X}}{\bar{Y}} (1 + \hat{x}_t - \hat{y}_t).$$

Since

$$\frac{X_t}{Y_t} = \frac{\bar{X}}{\bar{Y}} = C,$$

we have

$$C = C (1 + \hat{x}_t - \hat{y}_t) \quad \Rightarrow \quad \hat{x}_t - \hat{y}_t = 0.$$

(5)

$$X_t + Y_t = C \quad \Rightarrow \quad \bar{X} + \bar{Y} = C.$$

$$X_t + Y_t \approx \bar{X}(1 + \hat{x}_t) + \bar{Y}(1 + \hat{y}_t) = \bar{X} + \bar{Y} + \bar{X} \hat{x}_t + \bar{Y} \hat{y}_t$$

hence

$$\bar{X} \hat{x}_t + \bar{Y} \hat{y}_t = 0.$$

(6) budget constraint

$$C_t + K_t = Y_t + (1 - \delta)K_{t-1} \quad \Rightarrow \quad \bar{C} + \bar{K} = \bar{Y} + (1 - \delta)\bar{K}$$

$$C_t \approx \bar{C}(1 + \hat{c}_t), \quad K_t \approx \bar{K}(1 + \hat{k}_t), \quad Y_t \approx \bar{Y}(1 + \hat{y}_t), \quad K_{t-1} \approx \bar{K}(1 + \hat{k}_{t-1})$$

$$\Rightarrow \quad \bar{C}(1 + \hat{c}_t) + \bar{K}(1 + \hat{k}_t) = \bar{Y}(1 + \hat{y}_t) + (1 - \delta)\bar{K}(1 + \hat{k}_{t-1})$$

$$\Rightarrow \bar{C} + \bar{C}\hat{c}_t + \bar{K} + \bar{K}\hat{k}_t = \bar{Y} + \bar{Y}\hat{y}_t + (1 - \delta)\bar{K} + (1 - \delta)\bar{K}\hat{k}_{t-1}$$

Using

$$\bar{C} + \bar{K} = \bar{Y} + (1 - \delta)\bar{K},$$

we obtain

$$\bar{C}\hat{c}_t + \bar{K}\hat{k}_t = \bar{Y}\hat{y}_t + (1 - \delta)\bar{K}\hat{k}_{t-1}.$$

(7) Utility FOC

$$C_t^{-\sigma} = \lambda_t \quad \Rightarrow \quad \bar{C}^{-\sigma} = \bar{\lambda}$$

Let

$$C_t = \bar{C}e^{\hat{c}_t}, \quad \lambda_t = \bar{\lambda}e^{\hat{\lambda}_t}.$$

Then

$$C_t^{-\sigma} = \bar{C}^{-\sigma}e^{-\sigma\hat{c}_t} = \bar{\lambda}e^{\hat{\lambda}_t}.$$

Since $\bar{C}^{-\sigma} = \bar{\lambda}$,

$$e^{-\sigma\hat{c}_t} = e^{\hat{\lambda}_t}.$$

Using

$$e^{-\sigma\hat{c}_t} \approx 1 - \sigma\hat{c}_t, \quad e^{\hat{\lambda}_t} \approx 1 + \hat{\lambda}_t,$$

we get

$$1 - \sigma\hat{c}_t = 1 + \hat{\lambda}_t \quad \Rightarrow \quad -\sigma\hat{c}_t = \hat{\lambda}_t.$$

(8) C-D production function

$$Y_t = A_t K_t^\alpha N_t^{1-\alpha} \quad \Rightarrow \quad \bar{Y} = \bar{A} \bar{K}^\alpha \bar{N}^{1-\alpha}$$

$$\log Y_t = \log A_t + \alpha \log K_t + (1 - \alpha) \log N_t$$

$$\log \bar{Y} = \log \bar{A} + \alpha \log \bar{K} + (1 - \alpha) \log \bar{N}$$

Subtract:

$$\log Y_t - \log \bar{Y} = (\log A_t - \log \bar{A}) + \alpha(\log K_t - \log \bar{K}) + (1 - \alpha)(\log N_t - \log \bar{N})$$

hence

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_t + (1 - \alpha) \hat{n}_t.$$

(9) GDP identity

$$Y_t = C_t + I_t + G_t$$

From (5) we know

$$\hat{y}_t = \frac{\bar{C}}{\bar{Y}} \hat{c}_t + \frac{\bar{I}}{\bar{Y}} \hat{i}_t + \frac{\bar{G}}{\bar{Y}} \hat{g}_t = s_c \hat{c}_t + s_i \hat{i}_t + s_g \hat{g}_t.$$

share of GDP for each component

(10) Capital law of motion

$$K_{t+1} = (1 - \delta)K_t + I_t \quad \Rightarrow \quad \bar{K} = (1 - \delta)\bar{K} + \bar{I}$$

$$\Rightarrow \quad \bar{K}(1 + \hat{k}_{t+1}) = (1 - \delta)\bar{K}(1 + \hat{k}_t) + \bar{I}(1 + \hat{i}_t)$$

$$\Rightarrow \quad \bar{K} + \bar{K}\hat{k}_{t+1} = (1 - \delta)\bar{K} + (1 - \delta)\bar{K}\hat{k}_t + \bar{I} + \bar{I}\hat{i}_t$$

Using

$$\bar{K} = (1 - \delta)\bar{K} + \bar{I},$$

we get

$$\bar{K}\hat{k}_{t+1} = (1 - \delta)\bar{K}\hat{k}_t + \bar{I}\hat{i}_t.$$

Thus

$$\hat{k}_{t+1} = (1 - \delta)\hat{k}_t + \frac{\bar{I}}{\bar{K}}\hat{i}_t.$$

Since

$$\bar{K} = (1 - \delta)\bar{K} + \bar{I} \quad \Rightarrow \quad \frac{\bar{I}}{\bar{K}} = \delta,$$

we obtain

$$\hat{k}_{t+1} = (1 - \delta)\hat{k}_t + \delta\hat{i}_t.$$

(11) Euler equation

$$U'(C_t) = \beta E_t [U'(C_{t+1})R_{t+1}]$$

use CRRA utility:

$$U(C_t) = \frac{C_t^{1-\sigma}}{1-\sigma} \quad \Rightarrow \quad U'(C_t) = C_t^{-\sigma}$$

So

$$C_t^{-\sigma} = \beta E_t [C_{t+1}^{-\sigma} R_{t+1}].$$

At steady state:

$$\bar{C}^{-\sigma} = \beta \bar{C}^{-\sigma} \bar{R} \quad \Rightarrow \quad 1 = \beta \bar{R}.$$

Let

$$C_t = \bar{C}e^{\hat{c}_t}, \quad C_{t+1} = \bar{C}e^{\hat{c}_{t+1}}, \quad R_{t+1} = \bar{R}e^{\hat{r}_{t+1}}.$$

Then

$$\bar{C}^{-\sigma} e^{-\sigma \hat{c}_t} = \beta E_t [\bar{C}^{-\sigma} e^{-\sigma \hat{c}_{t+1}} \bar{R} e^{\hat{r}_{t+1}}].$$

First-order approximation:

$$\bar{C}^{-\sigma}(1 - \sigma \hat{c}_t) = \beta E_t [\bar{C}^{-\sigma} \bar{R}(1 - \sigma \hat{c}_{t+1})(1 + \hat{r}_{t+1})]$$

$$= \beta E_t \left[\bar{C}^{-\sigma} \bar{R} \left(1 + \hat{r}_{t+1} - \sigma \hat{c}_{t+1} - \underbrace{\sigma \hat{c}_{t+1} \hat{r}_{t+1}}_{\approx 0} \right) \right]$$

Using $1 = \beta \bar{R}$:

$$1 - \sigma \hat{c}_t = 1 + E_t(-\sigma \hat{c}_{t+1} + \hat{r}_{t+1}).$$

Hence

$$-\sigma \hat{c}_t = E_t(-\sigma \hat{c}_{t+1} + \hat{r}_{t+1})$$

so

$$\hat{c}_t = E_t(\hat{c}_{t+1}) - \frac{1}{\sigma} E_t(\hat{r}_{t+1}).$$

Log linearization cheat sheet for economics

$$(1) XY : \quad \hat{x} + \hat{y}$$

$$(2) X/Y : \quad \hat{x} - \hat{y}$$

$$(3) X^\alpha : \quad \alpha \hat{x}$$

$$(4) \prod_i X_i^{\alpha_i} : \quad \sum_i \alpha_i \hat{x}_i$$

$$(5) \sum_i X_i : \quad \sum_i \frac{\bar{X}_i}{\sum_j \bar{X}_j} \hat{x}_i$$

$$(6) K' = (1 - \delta)K + I : \quad \hat{k}' = (1 - \delta)\hat{k} + \delta \hat{i}$$

$$(7) \pi_t = \frac{P_t}{P_{t-1}} : \quad \hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1}$$

$$(8) \text{CRRA MU} : \quad \widehat{u'(c)} = -\sigma \hat{c}$$

$$(9) \text{CES index } X = \left[\sum_i w_i X_i^\rho \right]^{1/\rho} : \quad \hat{x} = \sum_i s_i \hat{x}_i$$

$$\text{with } s_i = \frac{w_i \bar{X}_i^\rho}{\sum_j w_j \bar{X}_j^\rho}$$